



ST BENEDICT'S

SUBJECT	Physics Science
GRADE	11
EXAMINER	Mr McCoy
MODERATOR	Mr Hilder & Mrs Govender
DATE	13 July 2022
MARKS	165 marks
DURATION	150 minutes

**MARK
TOTAL**

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LEARNER NAME

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 20 pages and a Data Sheet (Please detach).
2. Please check that your question paper is complete.
3. Read the questions carefully.
4. Answer ALL the questions on the question paper and hand it in at the end of the assessment. Remember to write your name and surname in the space provided above.
5. Use the data and formulae whenever necessary.
6. Show your working in all calculations.
7. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
8. Answers must be expressed in decimal format, not left as proper fractions.
9. Where appropriate, express answers to TWO decimal places.
10. It is in your own interest to write legibly and to present your work neatly.
11. TWO blank pages (pages 21-22) are included at the end of the paper. If you run out of space for a question, use these pages. Clearly indicate the question number of your answer should you use this extra space.

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Question	1	2	3	4	5	6	7	8
Mark								
Moderated Mark								
Question Total	16	15	17	15	20	27	26	29

Comment:

QUESTION 1 - MULTIPLE CHOICE

16 MARKS

Answer these questions on the multiple-choice answer sheet below. Make a cross (X) in the box corresponding to the letter that you consider to be correct.

A	B	☒	D
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Here the option C has been marked as an example

1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D
1.4	A	B	C	D
1.5	A	B	C	D
1.6	A	B	C	D
1.7	A	B	C	D
1.8	A	B	C	D

1.1 Which set of physical quantities consists only of scalar quantities? (2)

- A Velocity, speed and time.
- B Displacement, velocity and acceleration.
- C Time, distance and speed.
- D Acceleration, speed and distance.

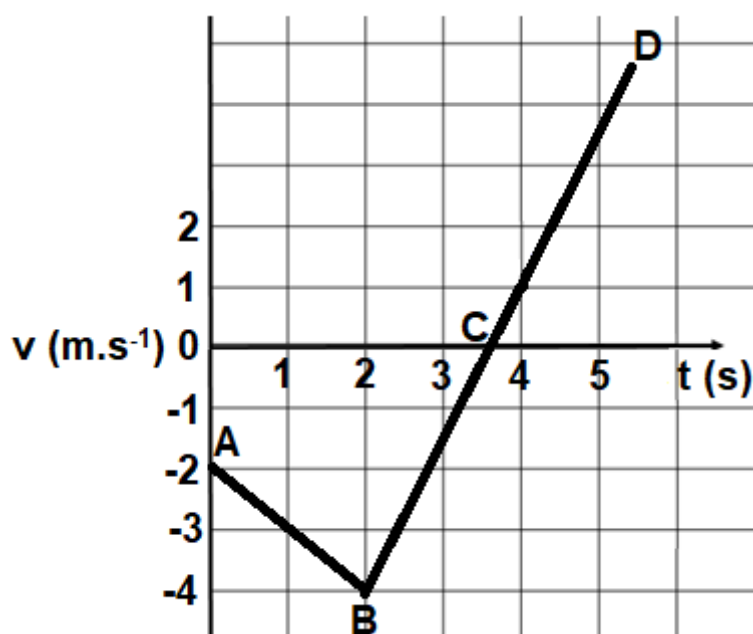
1.2 A ball is thrown vertically upwards and returns to the thrower's hand. Ignore air resistance. Which one of the following best describes the direction of the ball's **acceleration** during its flight? (2)

	Ball travelling upwards	Ball at top of flight	Ball travelling downwards
A	Downward	Downward	Downwards
B	Upward	Zero	Downwards
C	Downward	Zero	Downwards
D	Upward	Downward	Upwards

- 1.3 A ball thrown vertically upwards with an initial velocity v_i reaches a maximum height y . The velocity of the ball when it is halfway up is: (2)

- A $\frac{v_i}{2}$
 B $\sqrt{v_i - 2gy}$
 C gy
 D $\frac{1}{\sqrt{2}} v_i$

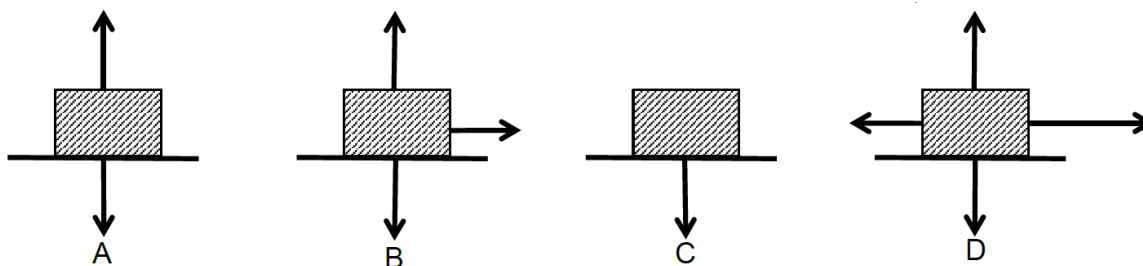
- 1.4 The following velocity vs time graph represents the motion of an object initially moving north. Refer to the graph and answer the questions that follow. At which point is the object the greatest distance from the origin? (2)



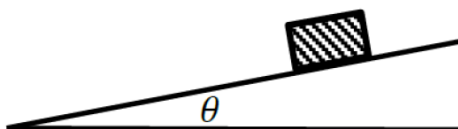
- 1.5 Which of the following would give a maximum resultant of vector **X** and vector **Y**? (2)

A	B
C	D

- 1.6 A block is moving with constant velocity to the right on a frictionless surface. Which sketch below correctly illustrates all the forces acting on the block? (2)



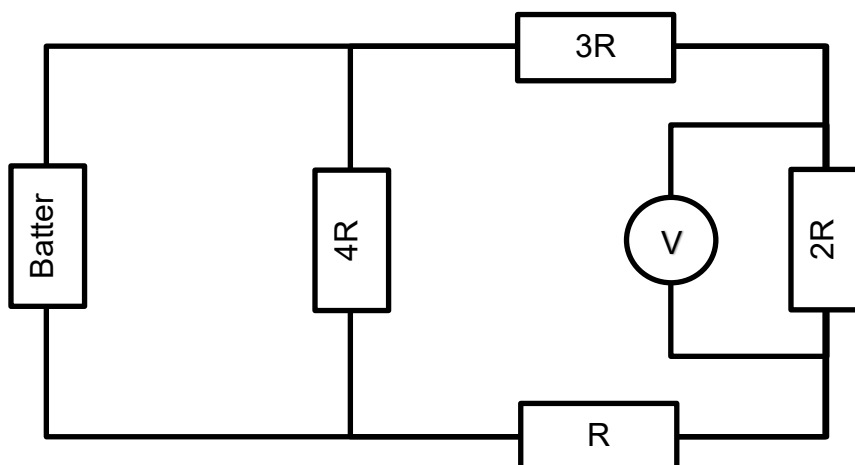
- 1.7 A block rests on an inclined rough surface that makes an angle θ with the horizontal.



The block remains at rest as θ is slowly increased. Which is the best description of how the magnitudes of the normal force and the static frictional force of the rough surface on the block change as θ is slowly increased? (2)

	Normal Force	Static Frictional Force
A	Increase	No change
B	Decreases	Decreases
C	Increases	Increases
D	Decreases	Increases

- 1.8 Four resistors of resistance R , $2R$, $3R$ and $4R$ are connected to form a network. A battery of negligible internal resistance and a voltmeter are connected to the resistor network as shown. The voltmeter reading is $2V$.



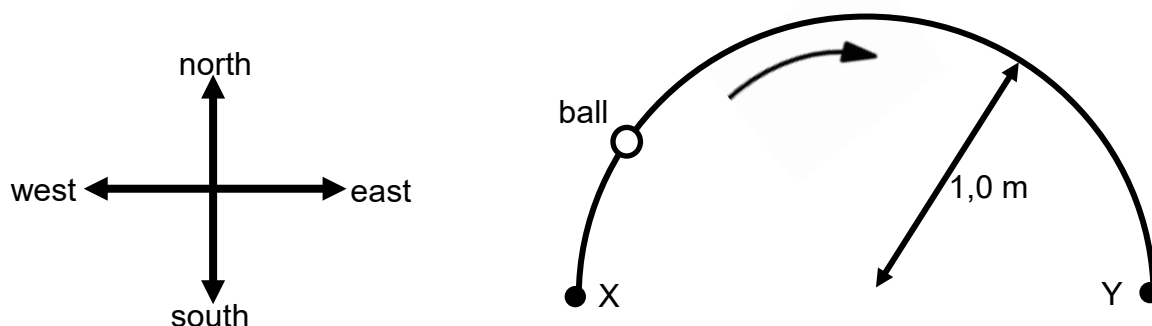
What is the emf of the battery? (2)

- A 2 V
- B 4 V
- C 6 V
- D 10 V

QUESTION 2

15 MARKS

A ball travels from Point X to Point Y at a constant speed along a horizontal semi-circular track of radius 1,0 m in 2 s as shown below.



2.1 Define the term *vector*. (2)

2.2 What is the displacement of the ball from X to Y? (2)

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2.3 Define *velocity*. (2)

2.4 Calculate the average velocity of the ball between X and Y. (3)

2.5 Determine the speed of the ball between X and Y. (4)

2.6 State the velocity of the ball at position Y. (2)

QUESTION 3

17 MARKS

A bouncing ball is thrown vertically upwards from the top of a building and falls to the ground below and bounces to a height of 9 m. The building is 18 m high.

If the ball reaches the ground in 2,8 s, calculate:

3.1 The magnitude of the initial velocity of the bouncing ball. (4)

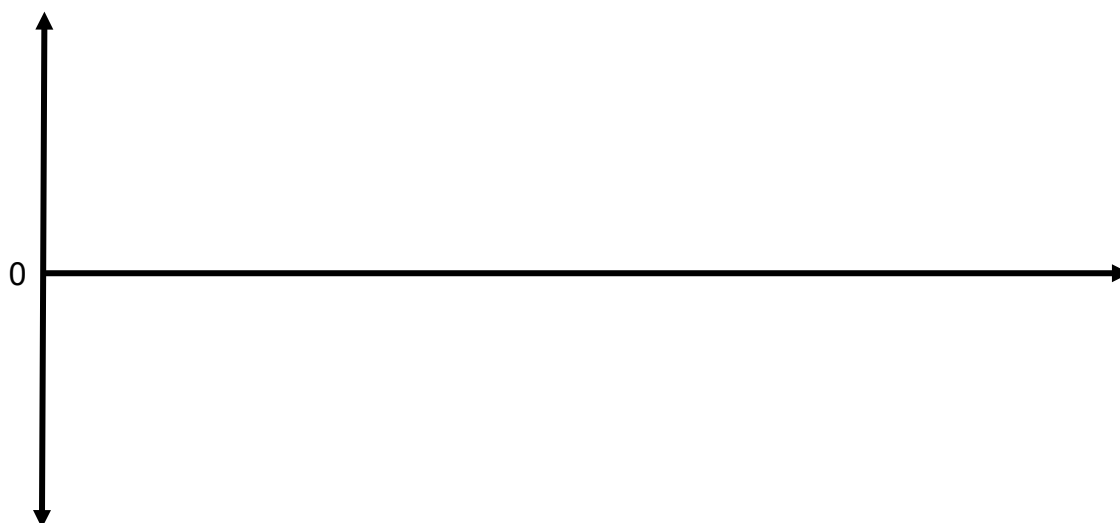
3.2 The maximum height the bouncing ball reaches above the **ground**. (4)

3.3 Sketch a labelled velocity-time graph of the bouncing ball's motion.

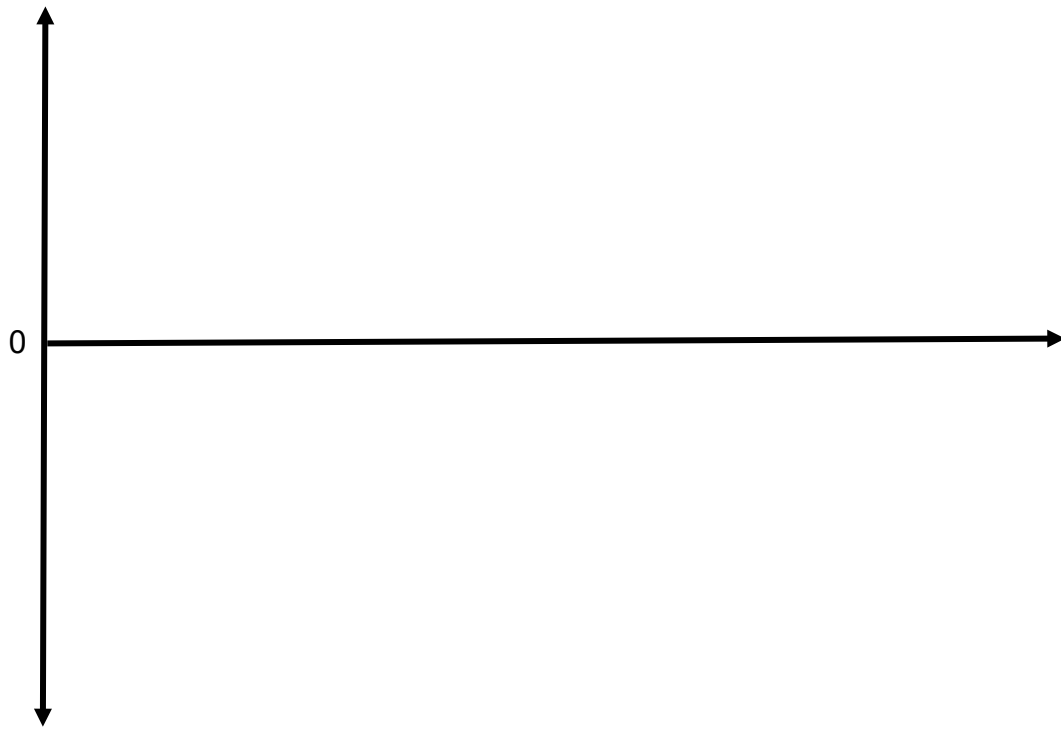
- Include values of initial velocity and time of flight.

- Label the point at which maximum height is reached.

(5)



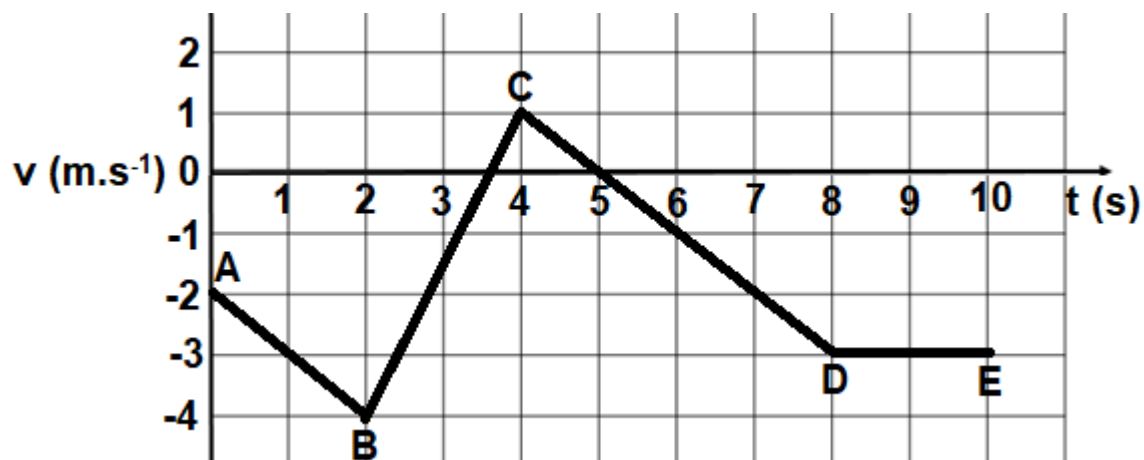
- 3.4 Sketch the corresponding displacement-time graph. Consider the top of the building as the position where the displacement is zero. (4)



QUESTION 4

15 MARKS

The following velocity vs time graph represents the motion of an object initially moving north. Refer to the graph and answer the questions that follow.



4.1 During which time points is the objects velocity constant. (2)

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4.2 Define *acceleration*. (2)

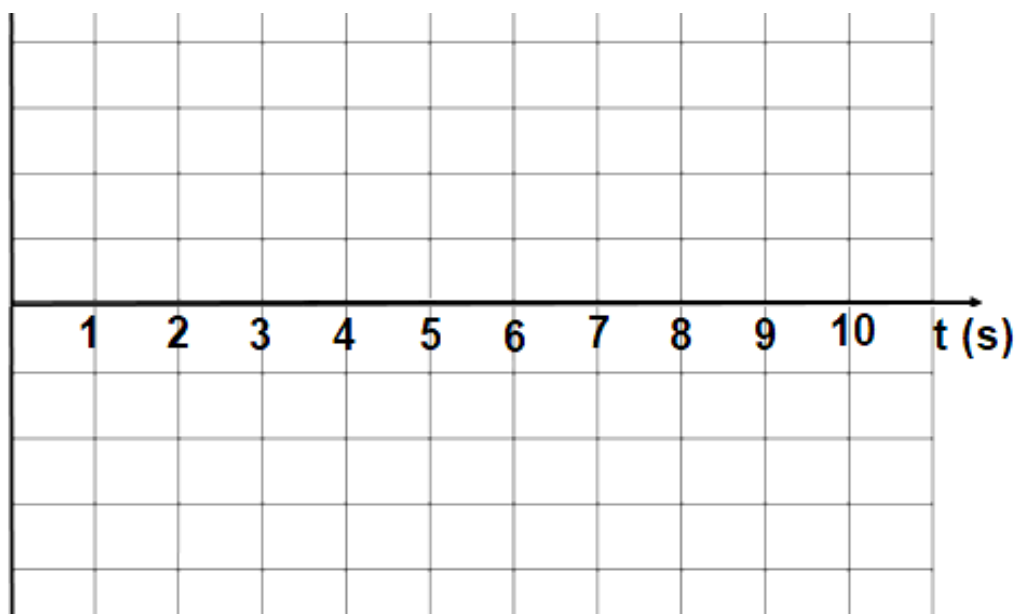
4.3 Without doing any calculations, compare the object's acceleration from AB to that of CD. Explain your answer with reference to the graph. (3)

4.4 How many times does the object change direction? (1)

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- 4.5 Calculate the distance covered by the object from B to C (indicate which values are used by adding dotted lines to the graph). (3)

- 4.6 Draw an acceleration vs time sketch graph for the motion. Label points A to E, axis values are not required. (4)

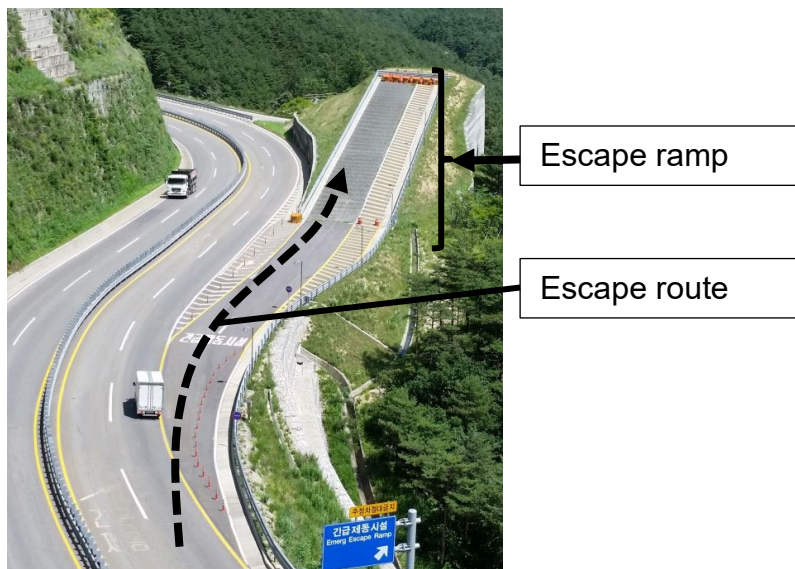


QUESTION 5

20 MARKS

A student was on holiday and saw an escape ramp for trucks. The ramp is used for trucks in an emergency that need to come to a **stop** when travelling downhill.

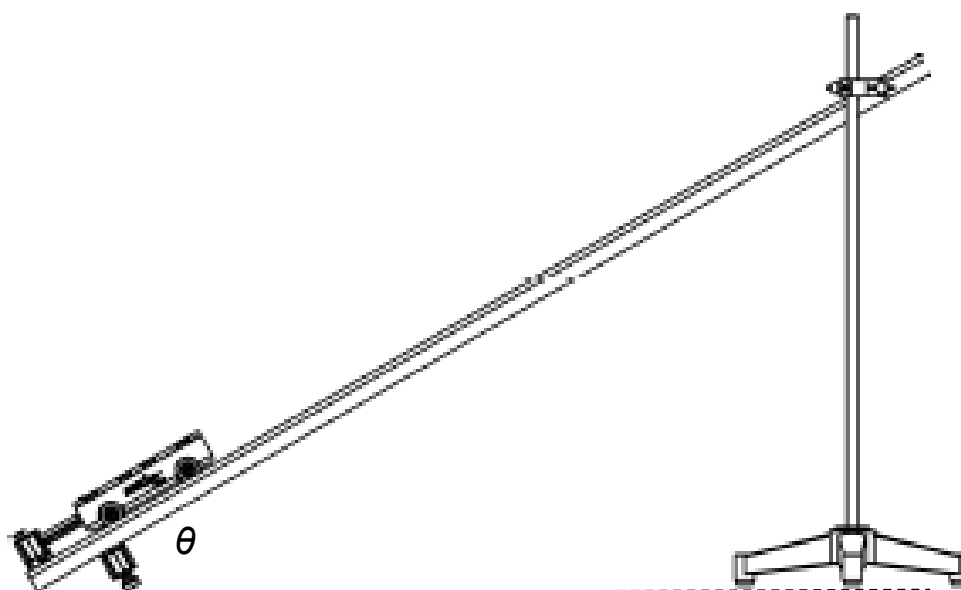
The student wonders if there is a relationship between the speed of the truck entering the ramp and the length of the ramp needed for the truck to stop.



5.1 Write a hypothesis for the investigation.

(2)

The investigation is recreated in the school laboratory using a Smart Cart, Chromebook and other equipment found in the school laboratory.



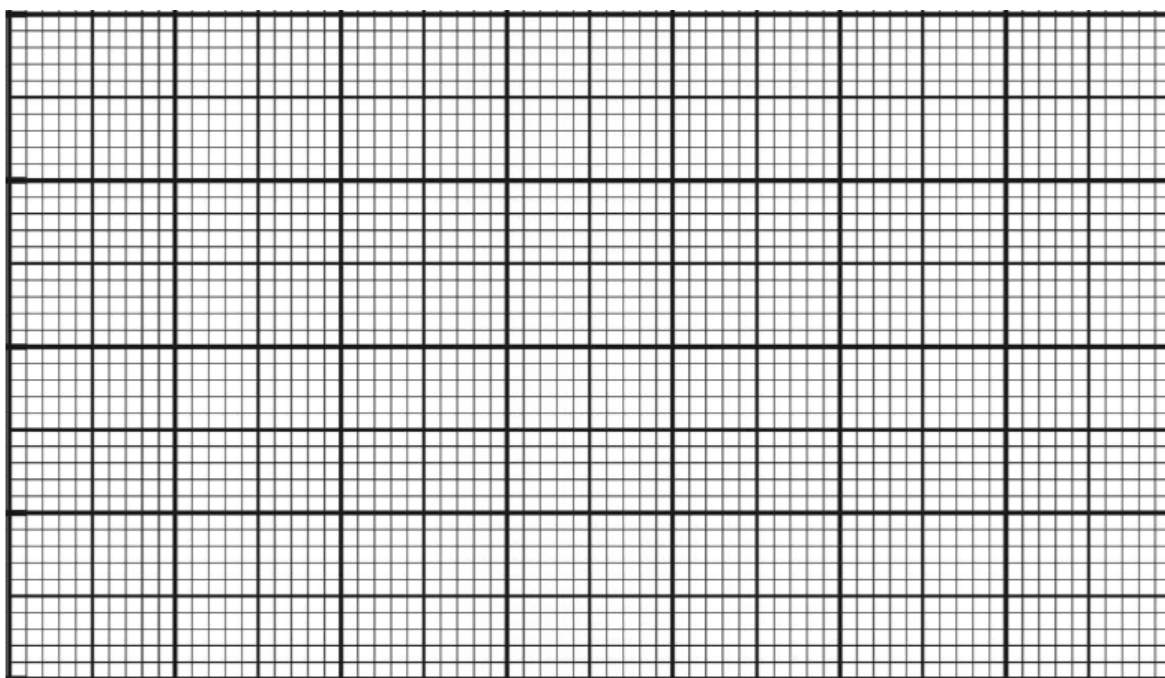
5.2 How could the student vary the initial velocity of the cart? (2)

Data is collected and placed in the table below.

Initial velocity of cart (m.s ⁻¹)	Average displacement (m)	Initial velocity ² (unit)
0,5	0,07	0,25
1,0	0,29	1,00
1,2	0,42	1,44
1,6	0,75	2,56
1,8	0,95	5.3)

5.3 Calculate the missing value in the table. (1)

5.4 Plot a graph of velocity² (on the y-axis) vs average displacement (on the x-axis) on the graph paper below. (6)



5.5 Draw a second line on the graph paper to show the effect of decreasing the angle θ . Label this line 5.5. (2)

5.6 Calculate the gradient of the graph. Include a unit.

(3)

5.7 Assume the gradient of the graph is **3.4**.

Use this value and your knowledge of equations of motion to determine the average acceleration of the cart.

(4)

QUESTION 6

27 MARKS

6.1 A robot moves 110 m N, then 150 m W, then 30 m S, then 100 m E, and lastly 100 m S.

6.1.1 Define *resultant vector*. (2)

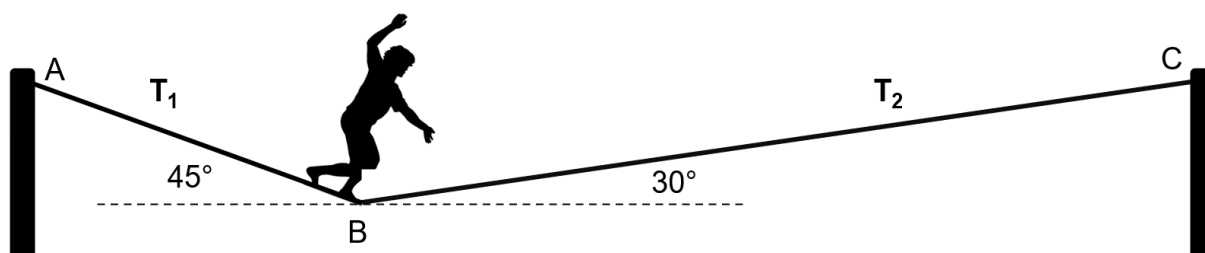
6.1.2 Calculate the magnitude of the north-south component. (1)

6.1.3 Calculate the magnitude of the east-west component. (1)

6.1.4 Use your answers from 6.1.2 and 6.1.3 to draw a fully labelled vector addition diagram to determine the resultant displacement from your starting point. (3)

6.1.5 Calculate the resultant displacement from your starting point. Give the answer using bearing. (4)

- 6.2 A 70 kg circus performer rests at point B on a rope ABC as shown below. BA and BC slope up from the horizontal at 45° and 30° respectively. BA experiences tension T_1 and BC experiences a tension of T_2 . The system is in equilibrium.



- 6.2.1 Define *weight*. (2)

- 6.2.2 Calculate the weight of the performer. (3)

- 6.2.3 What is the magnitude of the net force acting on the circus performer? (1)

- 6.2.4 Draw a free body diagram showing all the forces acting on the rope at **point B**. (3)

- 6.2.5 Write an equation relating the y-components of T_1 and T_2 to the mass of circus performer. (2)

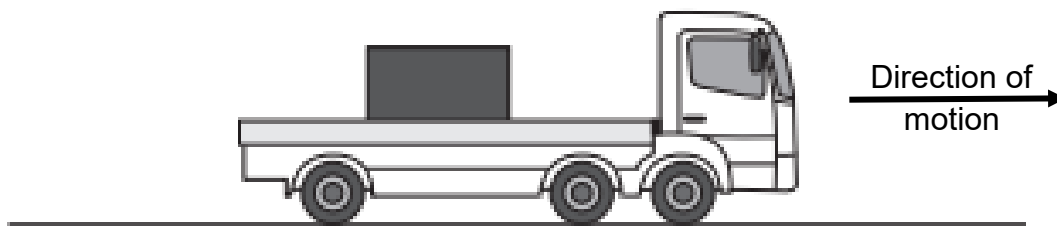
6.2.6 Calculate the magnitude of the tension in T_2 .

(5)

QUESTION 7

26 MARKS

There is a box of mass 200 kg on the back of a truck. The box is not tied to the truck.



7.1 Define *normal force*. (2)

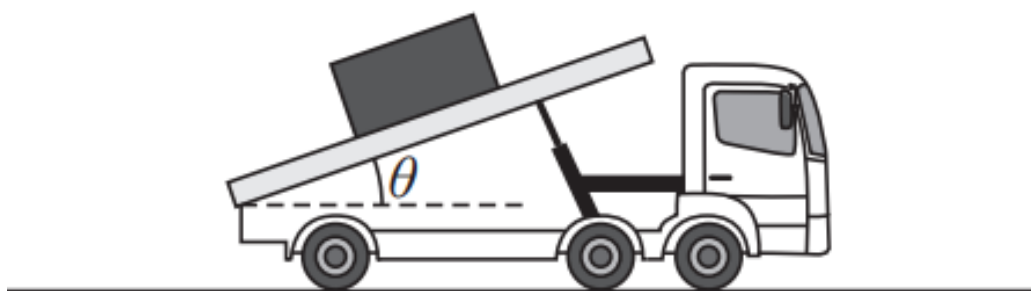
7.2 Define *friction due to a surface*. (2)

7.3 The maximum static frictional force between the truck and the box is 570 N.

7.3.1 What is meant by maximum static frictional force? (2)

7.3.2 Sketch a graph to show the relationship between the static friction and the acceleration of the truck. Fill in all known values. (3)

The truck has reached its destination, with the box still in its original position. The back of the truck is tilted at an angle θ to the horizontal.



7.4 Draw a labelled free-body diagram of all the forces acting on the box. (4)

7.5 State *Newton's third law*. (2)

7.6 Give the reaction force, with direction, to the weight of the box. (2)

7.7 The angle θ is increased until the box is just about to slide.

7.7.1 With reference to the force on the box; at what point will the box start to slide off the back of the truck. (2)

- 7.7.2 Calculate the angle θ that will cause the box to start sliding, if the coefficient of static friction between the box and truck is 0,32. Show **all** your calculations. (5)

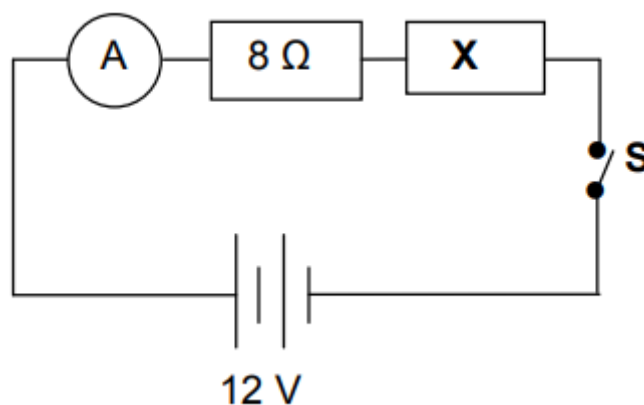
7.7.3 If the box starts to slide at $17,74^\circ$.

How will it be possible to reduce the angle that the box slides at? (2)

QUESTION 8

29 MARKS

8.1 Consider the circuit below which is used to determine the resistance of resistor **X**.

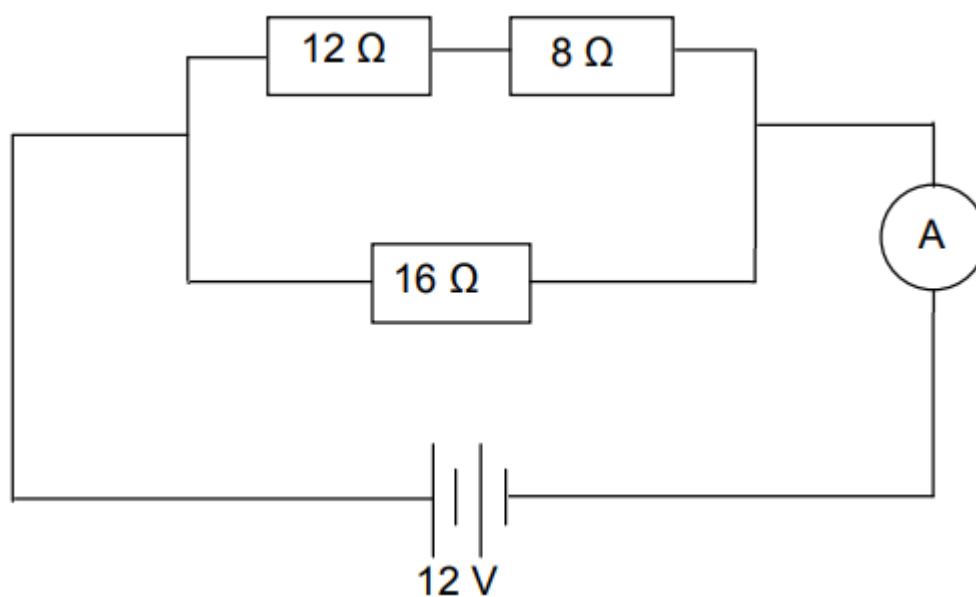


The 12 V battery has negligible internal resistance. When switch **S** is closed, the reading on the ammeter is 0,5 A.

- 8.1.1 State *Ohm's law* in words. (2)

8.1.2 Calculate the resistance of resistor **X**. (5)

3.2 Study the circuit below. The battery has an emf of 12 V with negligible internal resistance.



3.2.1 Define *resistance*. (2)

3.2.2 What is the potential difference across the 16 Ω resistor (2)

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3.2.3 Calculate the total resistance of the circuit. (4)

3.2.4 Calculate the reading on the ammeter. (3)

- 3.2.5 Calculate the energy dissipated in the $12\ \Omega$ resistor in 2 minutes. (6)

The $16\ \Omega$ resistor burns out.

- 3.2.6 How will the reading on ammeter A be affected? Write down only INCREASE, DECREASE or REMAINS THE SAME. (2)
- 3.2.7 Explain the answer to QUESTION 3.2.6 using a relevant equation. (3)

END OF ASSESSMENT

ACKNOWLEDGEMENTS

HILTON COLLEGE, GRADE 12 TRIALS EXAMINATIONS, AUGUST 2021, PAPER 1
ST JOHN'S COLLEGE, GRADE 12 PRELIMINARY EXAMINATIONS, AUGUST 2021, PAPER 1

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